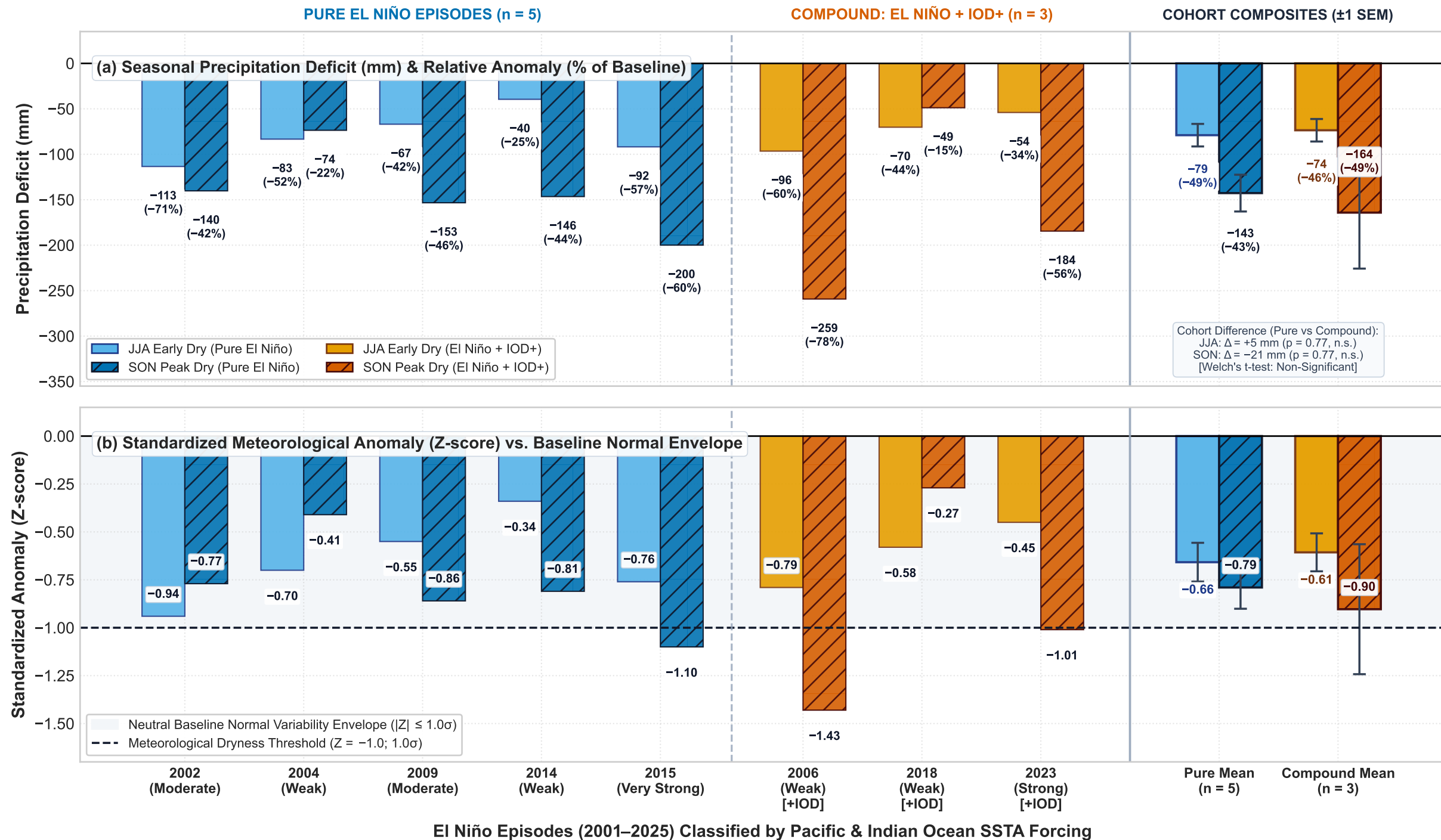


East Java Precipitation Response Across 8 El Niño Episodes (2001–2025)

Multi-Sensor Harmonized Domain | Neutral Baseline Climatology (n = 6): JJA = 161 mm, SON = 332 mm



• Data Source: UCSB CHIRPS v2.0 (native 0.05° ~5.5 km resolution). Baseline Climatology: 6 ENSO-neutral years (2001, 2003, 2012, 2013, 2019, 2025) within 2001–2025 multi-sensor domain.
• Criteria & Episodes: Pure El Niño (NOAA CPC ONI $\geq +0.5^{\circ}\text{C}$, ≥ 5 seasons); Compound IOD+ (BoM/NOAA DMI $\geq +0.4^{\circ}\text{C}$ in JJA/SON). 2014 & 2015 were consecutive years of a multi-year event.
• Uncertainty & Rigor: Error bars show ± 1 SEM. Cohort differences (Pure vs Compound) are statistically non-significant ($p = 0.77$). Baseline mean SE: JJA ± 49 mm, SON ± 74 mm; 2019 had extreme IOD+.